Digital Signal Processing

Relating Frequencies of Continuous and Discrete-Time Signals
An Introduction to Quantization

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www.lasse.ufpa.br & www.ppgee.ufpa.br

Remind: Periodic sampling

The sampled signal:

$$x_s(t) = \sum_{k=-\infty}^{\infty} x(kT_s)\delta(t - kT_s)$$

The sampled signal is given by periodically positioning an impulse whose area is the same as the amplitude of the original signal at the time it was sampled

The discrete-time signal:

$$x[n] = x(nT_s)$$

$$x[n] = \sum_{k=-\infty}^{\infty} x(kT_s)\delta[n - k]$$

The amplitudes of a discrete-time signal obtained from sampling periodically are the same of the original signal at the sampling times.

Is there a way to relate an analog-time ω(rad/s) or f (Hz) to a discrete-time Ω (rad)?

A portal between worlds

Suppose we are sampling a sinusoid $x(t) = A\cos(\omega t)$ at a sampling rate F_s, that is, sampling period $T_s = \frac{1}{F_s}$

$$x[n] = x(nT_s) = x\left(\frac{n}{F_s}\right)$$

$$x[n] = A\cos\left(\frac{\omega n}{F_s}\right)$$

Compare the above equation to a general discrete-time sinusoid:

$$x[n] = A\cos(\Omega \mathrm{n})$$

We then obtain $\Omega = \omega / F_s$

https://feedobem.com/as-cronicas-de-narnia/

For those who like analyzing dimensions: $\omega[rad/s]$, $\Omega[rad]$, $F_s[\mathrm{Hz\ or\ s^{-1}}]$

Examples ($F_s = 20$ Hz)

Continuous to Discrete

$x(t) = \cos(2t)$ ❯ $x[n] = \cos(0.1n)$

$x(t) = 2\sin(18\pi t)$ ❯ $x[n] = 2\sin(0.9\pi n)$

Discrete to Continuous

$x[n] = \cos(\pi n/4)$ ❯ $x(t) = \cos(5\pi t)$

$x[n] = \sin(0.7\pi n)$ ❯ $x(t) = \sin(14\pi t)$

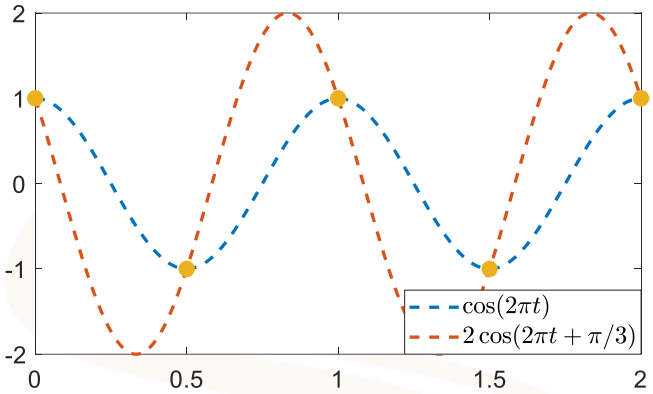
What happens if sampling theorem is not obeyed?

$$x(t) = \cos(2\pi 1t) \Rightarrow f_{max} = 1\text{Hz}$$

$F_s = 2\text{Hz}$

$$x[n] = \cos(\pi n) = (-1)^n$$

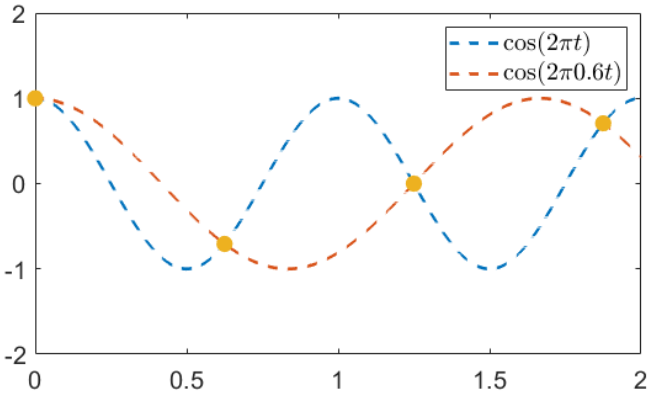
Loses phase and amplitude information!



$F_s = 1.6\text{Hz}$

$$x[n] = \cos(5\pi n/4)$$

Same result from:
$y(t) = \cos(2\pi 0.6t)$
$f_{\max} > \frac{F_s}{2}$ appears as a lower frequency

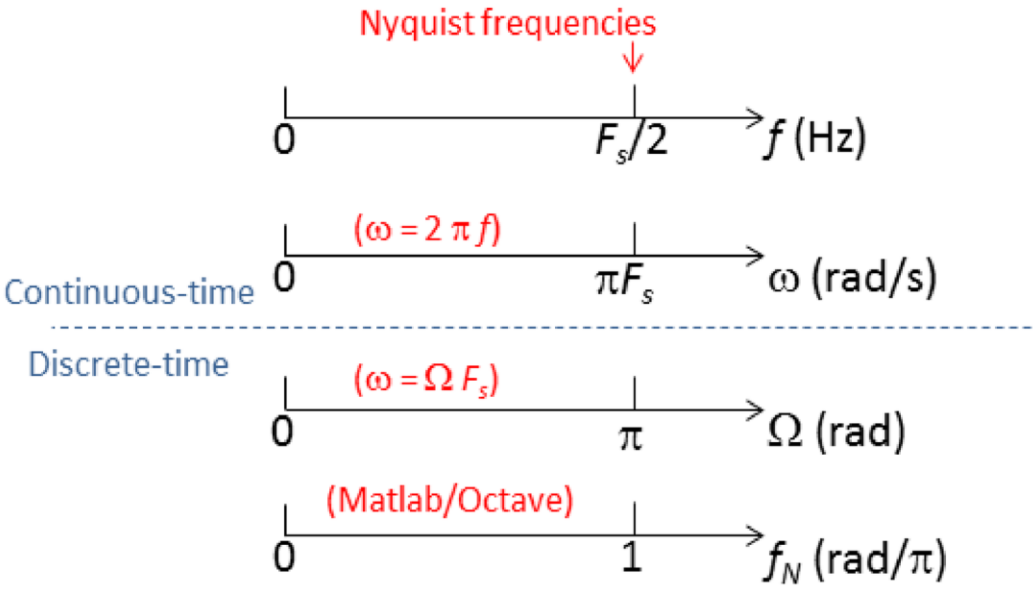


Frequency normalization

The abscissa interval $[0, \pi[$ is inconvenient to work with, so Matlab/Octave performs a normalization by dividing the axis by π

To relate the frequency in the linear form f with the final normalized frequency, you can use:

$$f_N = \frac{\Omega}{\pi} = \frac{\omega}{F_s \pi} = \frac{2\pi f}{F_s \pi} = \boxed{\frac{f}{F_s/2}}$$

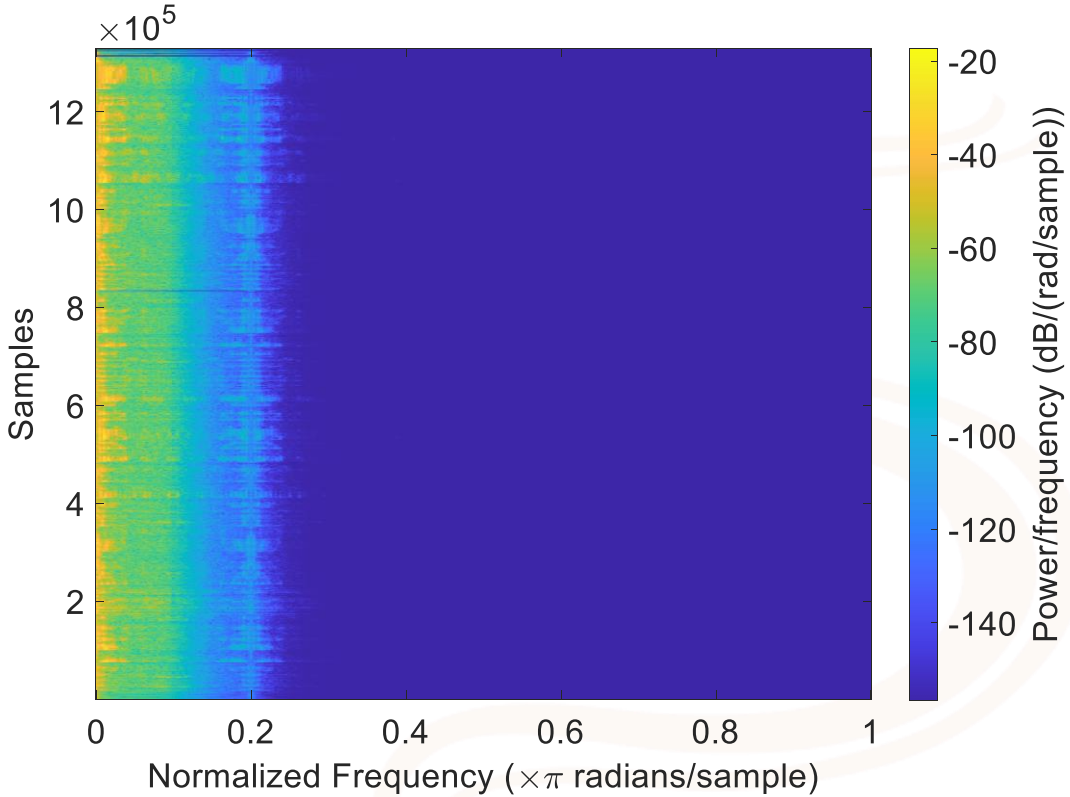


Using Spectrogram to estimate F_{max}

A spectrogram is used to analyze power distribution in frequency

A signal z[n] was sampled with $F_s = 80\text{kHz}$

The information is confined in lower frequencies. Can we decrease Fs?



The graph was generated with MATLAB: `w=hann(1000);spectrogram(z,w)`

Quiz-Time!

AD and DA frequency conversion

Normalized Frequency

Initial Definition

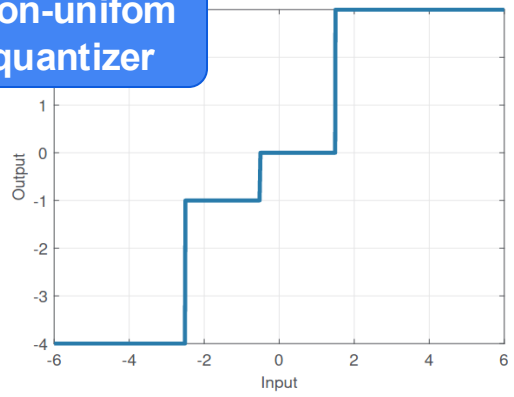
Converts the values from a discrete-time signal (x[n]) in a digital signal ($x_q[n]$)

$$x[n] \rightarrow \boxed{Q} \rightarrow x_q[n] \in \mathscr{M}$$

$M = 2^b$

Codebook: finite set with cardinality M

Non-unifom quantizer



$$\mathscr{M} = \{-4, -1, 0, 3\}$$

Using rounding

Input range	Output level
$]-\infty, -2.5[$	-4
$[-2.5, -0.5[$	-1
$[-0.5, 1.5[$	0
$[1.5, \infty[$	3

Uniform Quantizer

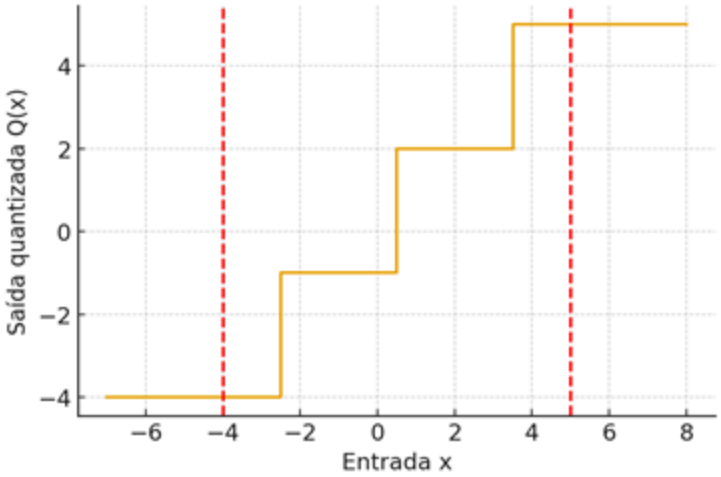
Defined by: number of bits (b), Δ (step size) and $\hat{X}_{\min}$

$$\mathscr{M} = \{\hat{X}_{\min}, \hat{X}_{\min} + \Delta, \hat{X}_{\min} + 2\Delta, \ldots, \hat{X}_{\min} + (M-1)\Delta\}$$

b=2

$\Delta = 3$

$\hat{X}_{\min} = -4$



$$\mathscr{M} = \{-4, -1, 2, 5\}$$

"Stairs" graph with same height and length, which are equals to a Δ (Step Size)

Granular and Overload Regions

Quantizer output Dynamic Range → $X_{min} \leftrightarrow X_{max}$

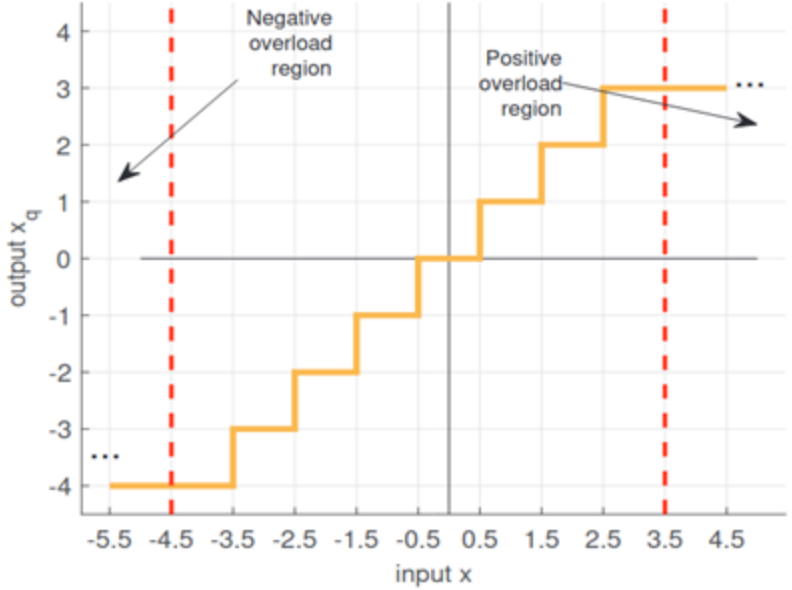


Figure 1.47: *Input/output relation of a 3-bits quantizer with* $\Delta = 1$.

Quantization error range ($e_q[n]$): $[-\Delta/2 , \Delta/2]$ within granular region

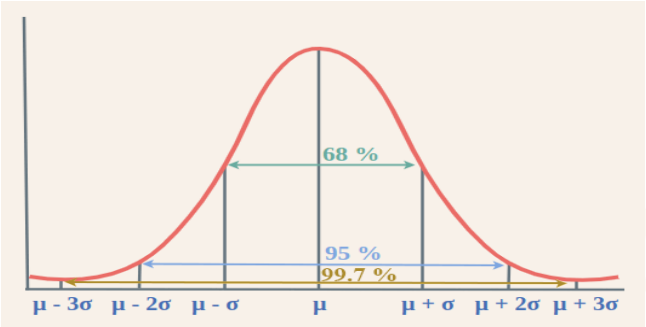
This error range only applies to a uniform quantizer, but the overload region applies to both uniform and non-uniform

Designing a Uniform Quantizer: input's statistics

There are several strategies for designing a uniform quantizer, statistics is one of them

If the input signal follows a Gaussian distribution, we can use its statistics (mean and variance) (μ and σ^2)

$X_{min} = \mu - 3\sigma$ $X_{max} = \mu + 3\sigma$ → Approximately 99% of samples



https://www.geeksforgeeks.org/machine-learning/gaussian-distribution-in-machine-learning/

Designing a Uniform Quantizer: Input is Dynamic Range

If you have the dynamic range of a quantizer, you can calculate the step size using:

Remind
$M = 2^b$

$$\Delta = \left| \frac{X_{\max} - X_{\min}}{M} \right| \quad \Rightarrow \quad \hat{X}_{\max} = X_{\max} - \Delta$$

$$\Delta = \left| \frac{X_{\max} - X_{\min}}{M - 1} \right| \quad \Rightarrow \quad \hat{X}_{\max} = X_{\max}$$

Designing a Uniform Quantizer: Input's Dynamic Range

Example from the main textbook to differentiate $\hat{X}_{\max}$ from $X_{\max}$

Example 1.39. **Design of a uniform quantizer.** For example, assume the quantizer should have $b = 2$ bits and the input $x[n]$ has a dynamic range given by $X_{\min} = -1$ V and $X_{\max} = 3$ V. Using Eq. (1.42) leads to $\Delta = (3 - (-1))/4 = 1$ V and the quantizer output levels are $\mathscr{M} = \{-1, 0, 1, 2\}$. Note that $\hat{X}_{\max} = X_{\max} - \Delta = 3 - 1 = 2$. Alternatively, Eq. (1.43) can be used to obtain $\Delta = (3 - (-1))/(4 - 1) \approx 1.33$ V and the quantizer output levels would be $\mathscr{M} = \{-1, 0.33, 1.66, 3\}$ with $\hat{X}_{\max} = X_{\max} = 3$. $\square$

Designing a uniform quantizer: input's dynamic range

One common requirement in a project of a quantizer is to reserve one level to 0, for this we can use:

$$X_{\min} + \Delta M_{\text{neg}} = 0$$

M_{neg} = number of negative values in the set $\mathcal{M}$

For example, using Δ=1.33

$$\mathcal{M} = \{-1, 0.33, 1.66, 3\} \quad \Rightarrow \quad \mathcal{M} = \{-1.33, 0, 1.33, 2.67\}$$

Designing a uniform quantizer: input's dynamic range

Another example: an 8-bits ADC can output signed integers from –128 to 127

-128 to 127 -> x_i values

$$\boxed{-2^{b-1}} = -2^{8-1} = -128$$

$$\boxed{2^{b-1} - 1} = 2^7 - 1 = 127$$

Then, to quantize the input value x to a x_q we do:

$$x_i = \text{round}\left(\frac{x}{\Delta}\right) \rightarrow x_q = x_i \times \Delta$$

Optimum quantizers

The uniform quantizer is optimum only when the input distribution is UNIFORM

If non-uniform distribution -> Lloyd's algorithm -> non-uniform quantizer

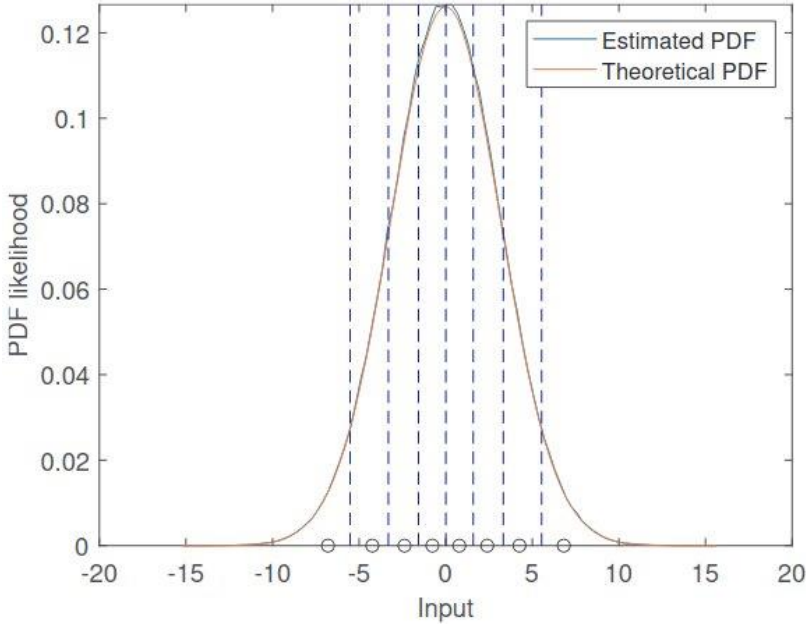
```
clf
N=1000000; %number of random samples
b=3; %number of bits
variance = 10;
x=sqrt(variance)*randn(1,N); %Gaussian samples
M=2^b;
[partition,codebook] = lloyds(x,M); %design the quantizer
```

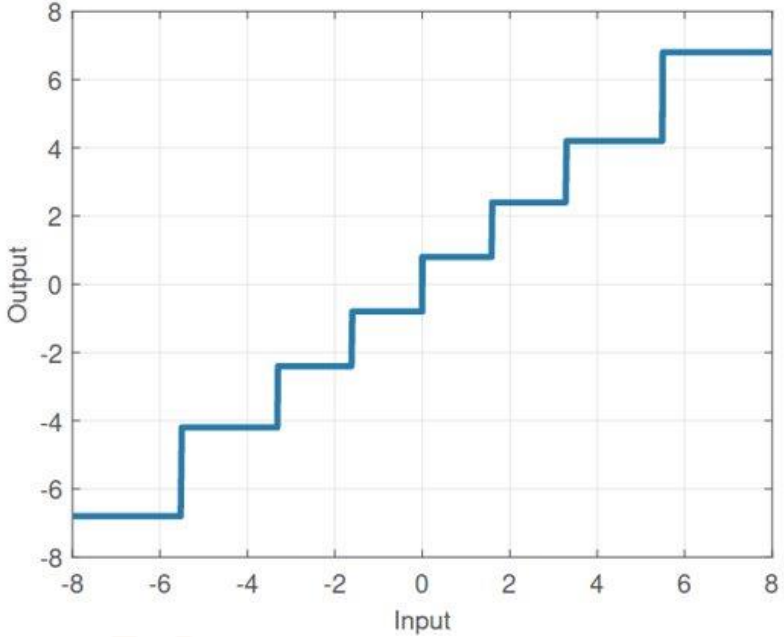
Example in MATLAB

You can find more information about the Lloyd's algorithm in https://en.wikipedia.org/wiki/Lloyd's_algorithm

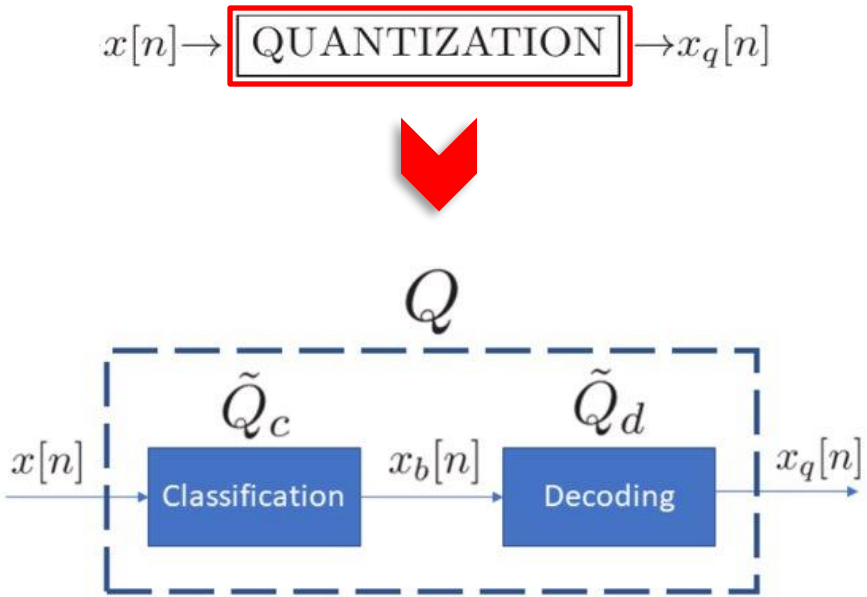
Optimum quantizers

Example of optimal quantizers. Note that there are more quantization levels in the area of highest density in the PDF





Quantization stages: classification and decoding



The decoding can be arbitrary, but, generally, a well-known binary numbering schemes are adopted

The decoding can be seen as the previous mentioned method to find x_q using x_i

$$x_q[n] = \tilde{Q}_d\{x_b[n]\} = x_i[n] \times \Delta$$

Quantization stages: classification and decoding

Table 1.8: *Examples of binary numbering schemes used as output codes in A/D conversion for* $b = 3$ *bits.*

Binary number	Unsigned integer $[0, 2^b - 1]$	Two's complement $[-2^{b-1}, 2^{b-1} - 1]$	Offset	Sign and magnitude	Gray code
000	0	0	−4	0	0
001	1	1	−3	1	1
010	2	2	−2	2	3
011	3	3	−1	3	2
100	4	−4	0	"−0"	7
101	5	−3	1	−1	6
110	6	−2	2	−2	4
111	7	−1	3	−3	5

Pay attention to headers and docs!

Binary: 100

Is this 4, -4, 0, -0 or 7?

https://knowyourmeme.com/memes/thinking-monkey

Programmers always need to check header files, documentation and datasheets to avoid errors when decoding binary data!

Table 1.8: *Examples of binary numbering schemes used as output codes in A/D conversion for $b = 3$ bits.*

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101	5	−3	1	−1	6
110	6	−2	2	−2	4
111	7	−1	3	−3	5

Quantization stages: classification and decoding

Example 1.45. **ECG quantization.** An ECG signal was obtained with a digital Holter that used a uniform quantizer with 8 bits represented in two's complement and quantization step $\Delta = 0.15$ mV. Hence, when represented as decimal numbers, the quantized outputs $x_i[n]$ vary from -128 to 127. Assuming three quantizer input samples as $x[0] = 2.52$, $x[1] = -14.8$ and $x[2] = 7.3$, all in mV, the quantized values are given by $x_i[n] = Q\{x[n]\} = \text{round}(x[n]/\Delta)$, which lead to $x_i[0] = 17$, $x_i[1] = -99$ and $x_i[2] = 49$. The corresponding binary codewords $x_b[n] = \tilde{Q}_c\{x[n]\}$ are $x_b[0] = 0001\ 0001$, $x_b[1] = 1001\ 1101$ and $x_b[2] = 0011\ 0001$. Using Eq. (1.46), the quantizer outputs are $x_q[0] = 2.55$, $x_q[1] = -14.85$ and $x_q[2] = 7.35$ mV, and the quantization error $x[n] - x_q[n]$ is $e_q[0] = -0.03$, $e_q[1] = 0.05$ and $e_q[2] = -0.05$ mV. □

Eq. (1.46):

$$x_q[n] = \tilde{Q}_d\{x_b[n]\} = x_i[n] \times \Delta$$

Number representation: Fixed vs. Floating Point

Fixed-point uses a specific value for the integer (b_i) and fractional (b_f) parts

Q3.4

$b_i = 3$ bits $b_f = 4$ bits

0	1	0	1	0	0	0	1
sign	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}

$$1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4} = 5.0625$$

Floating-point is defined as a scientific notation and can represent simultaneously small and large numbers

$$x_q = \text{significand} \times \text{base}^{\text{exponent}}$$

Exponent defines the position of the "point", unlike in the fixed case where the point is fixed

Problems of rounding

Precision (Fixed) X Range (Floating)

Quiz-Time!

Codebook

Uniform and non-uniform Quantization

Fixed vs Floating point representation

Thank you!

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